

Allison E. Grimsted
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EDUCATION

University of Michigan, College of Engineering, Ann Arbor, MI	August 2024 – Present
PhD, Industrial and Operations Engineering (IOE)	GPA: 3.91
<i>Advisor:</i> Professor Brian Denton	

Cornell University, College of Engineering, Ithaca, NY	August 2017 – May 2021
Bachelor of Science, Operations Research and Engineering (ORIE), <i>Magna Cum Laude</i>	GPA: 3.97
<i>Minors:</i> Mathematics; Biometry and Statistics	
<i>Research Advisor:</i> Professor Shane Henderson	

PROFESSIONAL EXPERIENCE

Johns Hopkins University Applied Physics Laboratory (JHU/APL)	Laurel, MD
<i>Associate Operations Research Analyst I</i>	August 2021 – July 2024
• Developed optimization algorithms, large-scale simulations, and data analytics pipelines to support complex operational decisions for federal agencies and internal research & development. • Led development and deployment of software for a federal agency, overseeing quality assurance and mentoring an intern. • Frequently briefed government program manager to demonstrate prototypes and display results. • Supported the federal COVID-19 response by building automated pipelines from numerous data streams into quality-controlled daily reports for government officials and public dissemination.	

AWARDS

<i>IOE Bonder Fellowship, University of Michigan</i>	2025
• Awarded competitive department fellowship, which provides full support for two semesters.	
<i>Coin Award, JHU/APL</i>	2022, 2023, 2024
• Awarded three times for leadership in research and development as an early career scientist. • 5% of JHU/APL technical staff win a coin award annually.	
<i>Building Leaders, Accelerating Science and Technology Grant, JHU/APL</i>	2022
• Awarded \$10,000 to fund portion of "Hybrid Bistatic Radar for In-Situ Debris Sensing." Less than 20% of grant applications awarded.	
<i>Nominee for Excellence in Quantitative Sciences, CDC</i>	2022
• Nominated for this award based on models and data analytics developed for the federal COVID-19 response in education.	
<i>Omega Rho, Cornell University</i>	2021
• Awarded to undergraduates in the top 15% of the ORIE major.	
<i>Engineering Learning Initiatives Undergraduate Research Grant, Cornell University</i>	2020
• Awarded \$4,500 to fund summer research, "Dispatch Policies for Urgent/Emergent Calls in Air and Land Ambulance Operations."	
<i>Tau Beta Pi, Cornell University</i>	2020
• Awarded to undergraduates in the top 12% of the engineering school.	

WORKING PAPERS

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1. **Grimsted, A.**, Orfanoudaki, A., Denton, B. (2026) *Sample-Path Clustering for Context-Sensitive Markov Decision Processes*.

TECHNICAL REPORTS

1. JHU/APL COVID-19 Operations Team (weekly 2021-2023). *COVID-19 State Profile Reports*, U.S. Department of Health and Human Services. <https://healthdata.gov/browse?tags=covid-19-spr>.
2. JHU/APL COVID-19 Operations Team (daily/weekly 2021-2023). *COVID-19 Community Profile Report*, U.S. Department of Health and Human Services. <https://healthdata.gov/Health/COVID-19-Community-Profile-Report/gqxm-d9w9>.
3. McGarity, M., Swan, B., Calloway, R., Baker, J., McShea, M., Beatty, P., **Grimsted, A.**, Hill, D., Howe, D., Jones, D., Matthew, K., Camacho, E., Brown, G., Llera Ortiz, J., Mekonnen, Y., Helmer, D. (2023). *Aeromedical Evacuation Human Performance Medical Mission Model Under Agile Combat Employment Environment – Summary Final Report*, JHU/APL Technical Communications, AOS-23-1509.
4. McGarity, M., Swan, B., Calloway, R., Baker, J., McShea, M., Beatty, P., **Grimsted, A.**, Hill, D., Howe, D., Jones, D., Matthew, K., Camacho, E., Brown, G., Llera Ortiz, J., Mekonnen, Y., Helmer, D. (2023). *Aeromedical Evacuation Human Performance Medical Mission Model Under Agile Combat Employment Environment – Analysis and Capability Gaps Assessment Report*, JHU/APL Technical Communications, AOS-23-1468.
5. **Grimsted, A.**, Frey, C. (2022). *Graph Based Analytic Model for Resilience Analysis*, JHU/APL Technical Communications, AOS-22-1208.
6. **Grimsted, A.**, Henderson, S. (2020). *Dispatch Policies for Urgent/Emergent Calls in Air and Land Ambulance Operations*, Engineering Learning Initiatives Report, Cornell University.

INVITED CONFERENCE PRESENTATIONS

1. **Grimsted, A.**, Orfanoudaki, A., Denton, B. (March 2026). *Sample-Path Clustering for Context-Sensitive Markov Decision Processes*, INFORMS Optimization Society Conference.
2. **Grimsted, A.**, Orfanoudaki, A., Denton, B. (2025). *Sample-Path Clustering for Context-Sensitive Markov Decision Processes*, INFORMS Annual Meeting.

INVITED SEMINAR PRESENTATIONS

1. **Grimsted, A.**, Swan Hovey, B. (2024). *Modeling Human Performance during a Simulated Aeromedical Evacuation Mission*, Modeling and Simulation Seminar, JHU/APL.
2. **Grimsted, A.** (2023). *Graph Based Analytic Model for Resilience Analysis*, Supply Chain Seminar, JHU/APL.

OTHER PRESENTATIONS

1. Flaherty, M., **Grimsted, A.**, Hingorani, R., Eddowes, D. (2022). *Hybrid Bistatic Radar for In-Situ Debris Sensing*, Building Leaders, Accelerating Science and Technology Capstone, JHU/APL.
2. **Grimsted, A.**, Maszkiewicz, L., Nichols, M., Raible, E., Gallagher, M. (2022). *National COVID-19 Response at JHU/APL*, Women in Technology Showcase, JHU/APL.
3. **Grimsted, A.**, Henderson, S. (2020). *Dispatch Policies for Urgent/Emergent Calls in Air and Land Ambulance Operations*, Engineering Learning Initiatives Summer Presentation, Cornell University.

POSTERS

1. Beatty, P., **Grimsted, A.** (2023). *Modeling Human Performance during a Simulated Aeromedical Evacuation Mission*, Asymmetric Operations Conference, JHU/APL.

2. **Grimsted, A.** (2022). *Graph Based Analytic Model for Resilience Analysis*, Asymmetric Operations Conference, JHU/APL.

INTELLECTUAL PROPERTY DISCLOSURES

1. Shin, L., **Grimsted, A.**, Behrendt, A. (2024). *Logistics Aid for Sustainment and Expeditionary Readiness*, PO7830.
2. McGarity, M., Swan, B., Calloway, R., Baker, J., McShea, M., Beatty, P., **Grimsted, A.**, Hill, D., Howe, D., Jones, D., Matthew, K., Camacho, E., Brown, G., Llera Ortiz, J., Mekonnen, Y., Helmer, D. (2023). *Human Performance Medical Mission Model (HPMx3)*, 7435-6400.

TEACHING

Teaching Assistant, ORIE 4741, Learning with Big Messy Data (Cornell University) Fall 2020

- Held weekly office hours to explain course content.
- Wrote solutions and grading guidelines for exams and homework.
- Managed a team of graders and resolved grade disputes.

Debate Coach (Evanston Township High School) Fall 2020 – Spring 2021

- Designed congressional debate curriculum for novice and experienced high school students during biweekly practices. Developed students' research skills.
- Judged at nation-wide tournaments and provided individualized feedback on students' speeches.

Grader, ORIE 3500/5500, Engineering Probability and Statistics II (Cornell University) Fall 2019

- Graded exams and homework for two sections.

SERVICE

Graduate Society of Women Engineers Ann Arbor, Michigan
Leadership Board Member, Activities Committee May 2025 – Present

- Organize social events for women in engineering graduate programs twice per month.

Engage Detroit Detroit, Michigan
Volunteer, Optimization Team June 2025 – July 2025

- Supported multi-day outreach workshops introducing optimization to Detroit high school students through real-world examples and interactive games.

IOE Graduate Application Mentoring Program Ann Arbor, Michigan
Mentor September 2024 – Present

- Provide application guidance and answer questions about the department graduate program for prospective students to the IOE PhD program.

Alpha Phi Omega – National Community Service Fraternity Ithaca, NY
Treasurer December 2018 – December 2019
Executive Board Member May 2018 – May 2021

- Managed finances for an organization with an \$80K+ annual budget and 550+ members.
- Created and began implementation of 2-year plan to decrease dues by more than 20%.
- Collected and analyzed survey data to improve chapter outcomes. Utilizing 10+ years of chapter data, deployed predictive analytics to target members at risk of not completing service hours to increase retention.
- Volunteered with Prisoner Express, an organization dedicated to addressing the isolation of incarcerated individuals by implementing educational initiatives for self expression such as creative writing classes and book curation.